

NON-MEASURABLE SETS.

We have $([0,1], \mathcal{F}, \lambda)$ a probability space s.t.

- $\mathcal{F} \supseteq \mathcal{B}$ (the Borel σ -alg.)
- $\lambda(I) = |I|$, $\forall$ intervals I .

Proposition $\mathcal{F} \subsetneq 2^{[0,1]}$, i.e., there exists subsets of $[0,1]$ which are not Lebesgue m'sble.

Setting/Notations

- Instead of $[0,1]$, we will consider $\underbrace{[0,1]}_{\sim}$ with the group operation $\oplus$

$$x \oplus y = x + y \pmod{1}$$
 and by a slight abuse of notation use $+$ instead of $\oplus$.

- This makes $[0,1]$ into a group.

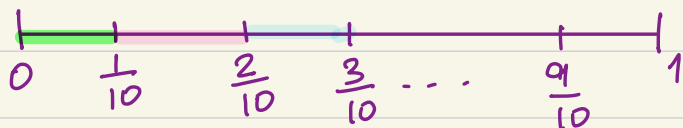
- Alternatively, one could think of $\mathbb{S}^1 = \{e^{2\pi i \theta} : \theta \in [0,1]\}$.

Exercise: For any $A \subseteq [0,1]$ & $x \in [0,1]$:

$$\lambda_*(A) = \lambda_*(A \oplus x).$$

Before proving the prop., here is an idea of the proof...

- Suppose we want to break $[0,1]$ into 10 "equal" pieces, then there is a natural way to do it:



$$[0,1] = [0, \frac{1}{10}] \cup [\frac{1}{10}, \frac{2}{10}] \cup \dots \cup [\frac{9}{10}, 1]$$

* The pieces are equal in the sense that each of them is a translate of the other.

* Also $\lambda_*(\text{each piece}) = \frac{1}{10}$.

- Q: What if we want to break $[0,1]$ into countably infinite "equal" pieces?

Axiom of Choice makes this possible!

Axiom of Choice:

Suppose $\{F_i\}_{i \in I}$ is a collection of non-empty sets, so each $F_i \neq \emptyset$.

Then $\exists$ a "choice function":

$$\varphi: I \rightarrow \bigcup F_i \text{ satisfying} \\ \varphi(i) \in F_i, \forall i.$$

$$\text{So, } [0,1] = A_0 \cup A_1 \cup A_2 \cup \dots$$

$$\text{where } A_i = A_0 + g_i.$$

So, A_i 's are "equal"/similar in the above sense. If the A_i 's were measurable, we

would have $\lambda(\underbrace{[0,1)}_1) = \sum_{i=1}^{\infty} \lambda(A_i)$,

but each $\lambda(A_i)$ is the same !!

Thus you see that we run into some problem...

Proof of Proposition:

- Let $Q = \mathbb{Q} \cap [0,1)$ This is a subgroup of $([0,1), \oplus)$.

- Consider $[0,1)/_Q = [0,1)/_{\sim}$ where

$\sim$ is the equiv. relation $x \sim y$ iff $x - y \in Q$.

- Use AC to "pick" an element from each coset of $[0,1)/_Q$ and form the set $A \subseteq [0,1)$.

[Recall if $H \subseteq G$ subgp, then consider the equiv

rel. on G : $g_1 \sim g_2 \Leftrightarrow g_2 g_1^{-1} \in H$.

• As a set $\frac{G}{H} :=$ Coll. of equiv. classes of G wrt $\sim$.

• There is a natural group structure on G/H } $[g_1] \cdot [g_2] = [g_1 g_2]$

• Every equiv. class is $g \cdot H$ for some $g \in G$

• Make a choice of representative g_α for each equivalence class & form $\{g_\alpha\}_{\alpha \in A}$.

Then $G = \sqcup g_\alpha H$.

]

Thus $[0,1) = \sqcup_{r \in \mathbb{Q}} A \oplus r$.

Suppose $A \in \mathcal{F}$, then b/c λ is a measure on $\mathcal{F}$, countable additivity would imply

$$1 = \lambda([0,1]) = \sum_{r \in \mathbb{Q}} \lambda(A \oplus r) \rightarrow (*)$$

$$\left. \begin{array}{l} \text{If } \lambda(A) = 0, \text{ then } (*) \Rightarrow 1 = 0 \\ \text{If } \lambda(A) > 0, \text{ then } (*) \Rightarrow 1 = \infty \end{array} \right\} (\Rightarrow \Leftarrow)$$

Thus $A \notin \mathcal{F}$ & is NOT measurable.
